

To: Technical Manager
From: Brandon Acosta
RE: IMDB Sentiment Classification Model - Deployment Recommendation

Summary

I developed a text classification model using a sample dataset of 50,000 IMDB movie reviews to determine whether each review is positive or negative. Two vectorization models, CountVectorizer and TF-IDF, and two algorithms, Multinomial Naive Bayes and Logistic Regression, were tested on the dataset and compared in terms of performance. Based on the final results, I recommend using **Logistic Regression with TF-IDF** as the deployment model.

What I Built

I used a database of 50,000 IMDB reviews, containing 25,000 positive and 25,000 negative reviews. Having a balanced dataset helped ensure models would perform up to standard. HTML formatting, extra spaces, or other non-text characters were cleaned before modeling. Punctuation, capitalization, and contractions were kept to ensure review sentiment was taken into consideration (i.e., phrases like “didn’t work”, “!!!”, and “NOT good” are all important). A total of 4 models were tested, with an 80/20 split of the 50k reviews.

Model Recommendation

I recommend Logistic Regression + Tfidf, based on the following metrics:

Recommended model: Logistic Regression + TF-IDF
F1 score: 0.8879
Negative recall: 0.8836
Speed (seconds): 0.3236

In our business setting, missed negative reviews are more costly than missed positive reviews. Viewer dissatisfaction reveals greater insights and constructive criticism for future movie production, rather than positive reviews. Additionally, positive reviews can be more shallow and subject to biases, favoritism, and fanaticism.

Business Implications

From a deployment perspective, this model would be useful for automatically sorting large volumes of movie reviews into positive and negative buckets. With this model, we could:

- Identify trends in audience dissatisfaction based on movie characteristics
- Monitor reaction to the latest movie releases and sequels
- Prioritize negative feedback for review of directors, casting, and marketing budgets

Key Limitation

This model is less effective for mixed, subtle, or extremely long reviews. Reviews that contain a mix of praise and criticism and sarcastic tones could make it harder for the model to classify sentiment accurately. To address this, I recommend (a) filtering out potentially lengthy or overly sarcastic reviews using other data preprocessing methods and/or (b) periodically retraining the model on fresh review data, especially those with complex language.